

Task 3 Contribution Journal — Feature Scaling

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My assigned part

My individual contribution to the Module 05 Data Preparation lab is Task 3: scaling the Age and Fare columns. I worked on applying StandardScaler from scikit-learn to the two numerical features in the prepared Titanic dataset.

Code and verification

I prepared the code to create a StandardScaler, select the Age and Fare columns, and transform those columns. I kept the target variable and the one-hot encoded categorical columns out of the scaling step, as instructed in the lab.

I included checks to display the first few rows after scaling, calculate the means, calculate population standard deviations, and count missing values in the scaled columns. The expected results are means close to zero, standard deviations close to one, and no missing values in these two columns after the earlier imputation step.

What this contribution demonstrates

This task demonstrates my understanding of how standardization puts numerical features on a comparable scale and how verification checks can be used to evaluate a data-preparation step. It also connects my task to the earlier workflow stages because the scaling code uses the encoded dataset created in Task 2.

Before submitting, I will run the notebook in Google Colab and review the actual outputs. The dataset URL was not reachable in the preparation environment, so I will only state that the live verification passed after I have run the cells and confirmed the results.